

Solution Code - JUnit, Mockito and SL4J - SLF4J_Logging_Logback

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: src\main\java\com\example\LoggingDemo.java

```
1 | package com.example;
2 |
3 | import org.slf4j.Logger;
4 | import org.slf4j.LoggerFactory;
5 |
6 | public class LoggingDemo {
7 |     private static final Logger logger = LoggerFactory.getLogger(LoggingDemo.class);
8 |
9 |     public static void main(String[] args) {
10 |         logger.info("Initializing application demo...");
11 |         try {
12 |             int result = 10 / 0;
13 |         } catch (ArithmeticException e) {
14 |             logger.error("Arithmetic exception caught: {}", e.getMessage(), e);
15 |         }
16 |     }
17 | }
```

Source File: src\test\java\com\example\AppTest.java

```
1 | package com.example;
2 |
3 | import org.slf4j.Logger;
4 | import org.slf4j.LoggerFactory;
5 |
6 | public class LoggingExample {
7 |     private static final Logger logger = LoggerFactory.getLogger(LoggingExample.class);
8 |     public static void main(String[] args) {
9 |         logger.error("This is an error message");
10 |         logger.warn("This is a warning message");
11 |         logger.info("This is an info message with parameter: {}", "TestValue");
12 |     }
13 | }
```